

EF5342 Financial Systems, Markets and Instruments

Week 7 — Foreign Exchange Markets

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The Foreign Exchange Market

Outline: overview of FX markets; quotes and trades; hedging FX risk; how exchange rates move; the carry trade and other currency strategies. The FX market's three functions:

Currency conversion Facilitates conversion of one country's currency into another — global firms move in and out of foreign currency as needed

Setting exchange rates The ratio of one currency to another; these rates determine costs and returns to global businesses

Hedging contracts Allow firms to offset foreign-currency exposure and manage FX risk — so they can concentrate on their core business

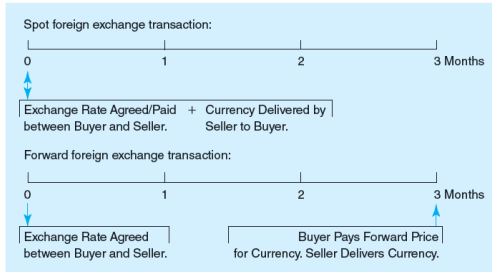
Spot and Forward Transactions

FX markets are where one currency is exchanged for another, today or at a set future date:

Spot Immediate exchange at current rates — ~27% of global FX turnover

Forward Exchange at a specified rate on a specific future date — 19% of turnover

FX swaps A spot leg plus an opposite forward leg, usually identical in notional — 42% of turnover



Market Participants

Four tiers by information advantage, motivation, and market access:

Tier	Participants	Primary motivation
1 (Core)	Major dealer banks (JPMorgan, UBS, Deutsche, Citi, Barclays)	Market-making, proprietary trading
2 (Interbank)	Regional and smaller banks	Customer service, hedging, arbitrage
3 (Buy-side)	Hedge funds, asset managers, pension funds	Speculation, portfolio hedging, carry/momentum
4 (End-users)	Corporations, central banks, retail	Transactional hedging, reserve management, policy

Global banks act for external clients (exporters, importers, multinationals) in a *broker* capacity, and for themselves in a *dealer* capacity. Through this market-maker function they set the “tone” of the market.

The Dollar's Long Run: DXY by Decade

Decade	Approx. average level	Trend	Dominant regime
1980s	~125–140	Peak & crash	Super-strong USD → Plaza Accord
1990s	~95–105	Mild uptrend	Strong-dollar era
2000s	~85–95	Downtrend	Weak USD, commodity supercycle
2010s	~85–100	Uptrend	Post-crisis recovery
2020s	~90–112	Volatile	COVID slump → tightening surge → flat near 99

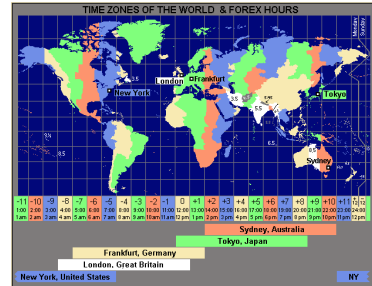
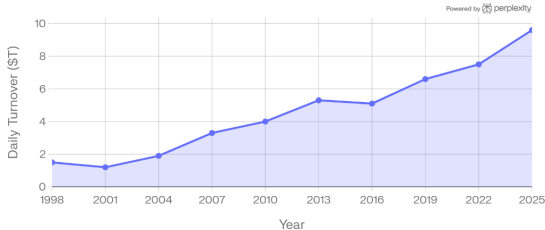
The Largest of All Financial Markets

FX markets facilitate international trade, raising capital abroad, transfer of risk, and speculation.

- Global FX turnover averaged **\$9.6 trillion per day** (BIS Triennial Survey, April 2025)
- Versus roughly \$80–300 billion per day on the NYSE

Global FX Market Growth 1998–2025 (Daily Turnover, \$ Trillion)

Source: BIS Triennial Central Bank Survey | 500%+ growth over 25 years



Stylized Facts

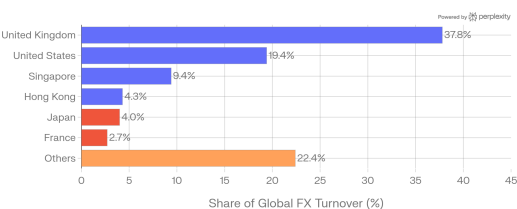
- Most traded currencies: **USD, EUR, JPY, GBP**; most traded pairs: USD per EUR, JPY per USD, USD per GBP, USD per AUD
- Counterparties: mostly *financial institutions* trading, not companies
- Types of transactions: spots (transact now, deliver now); forwards (transact now, deliver later); FX swaps (paired transactions, e.g., spot + forward)
- Horizons: forwards and swaps are mainly short-dated (~ 7 days) — mainly short-term speculation

Source: BIS Triennial Central Bank Survey (latest: April 2025; next: April 2028)

Where FX Trades, and What Trades Most

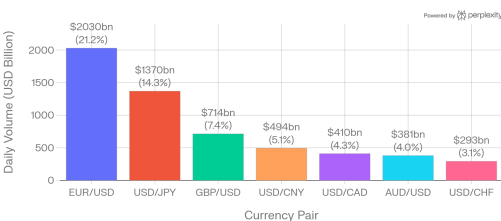
Geographic Concentration of Global FX Trading (April 2025)

Source: BIS Triennial Survey 2025 | Top 4 centres = 75% of total volume



Top Currency Pairs — Daily Trading Volume (April 2025)

Source: BIS Triennial Survey 2025 & BestBrokers | EUR/USD dominates at \$2.03T/day

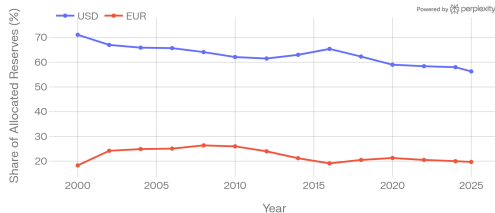


Top FX trading centers and top currency pairs — Source: BIS Triennial Central Bank Survey (April 2025; next 2028)

Reserves and Instruments

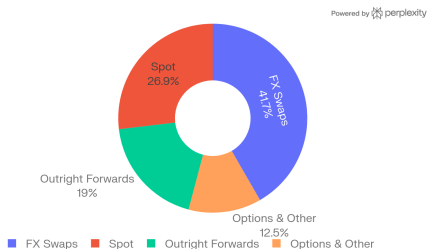
USD & EUR Share of Global Official FX Reserves (2000–2025)

Source: IMF COFER | USD at record-low 56.3% in mid-2025; down from 71% in 2000



Global FX Daily Turnover by Instrument (April 2025)

Total: \$9.6 Trillion/Day | Source: BIS Triennial Survey 2025



USD & EUR share of global FX reserves; FX turnover by instrument — Source: BIS Triennial Central Bank Survey (April 2025)

Dollar Dominance — and Renminbi Internationalization

- The USD is involved in ~**88%** of all FX transactions (one side) — unchanged for over two decades; deep network effects (Thailand buying Brazilian cotton typically goes through USD)
- But the dollar's share of allocated reserves has fallen from over 70% in 2000 to roughly **57%**, its lowest since the mid-1990s (IMF COFER, Q1 2026): euro ~20%, yen ~5%, pound ~5%, RMB ~2%
- **RMB internationalization** since 2009: cross-border trade settlement, bilateral swap lines with 30+ countries, CIPS, IMF SDR inclusion (2016) — yet still far behind USD, EUR, JPY
- **2025–26 tariff cycle**: US tariffs on China peaked at 145% in April 2025, and the Supreme Court struck down the IEEPA tariffs in February 2026 — yet the renminbi *appreciated*, from 7.4 to 6.8 per dollar. Why would a currency strengthen into a trade war?
- The fundamental constraint is **capital controls**: without free convertibility and deep, open capital markets, the RMB cannot be a genuine reserve asset

Types of Quotes

Market convention: all currencies quoted in *foreign currency per USD* (**European terms**) — except **EUR, GBP, AUD, NZD**, quoted in American terms.

Direct quote Home currency per unit of foreign currency — for a U.S. firm: 0.14 USD per CNY

Indirect quote Foreign currency per unit of home currency — 6.94 CNY per USD

European terms Foreign currency per USD; **American terms:** USD per unit of foreign currency

90 FX Markets Overview	
Base Currency	USD
FX Rate vs USD	DMMV »
	Spot
USD	1.0000
EUR	1.0783
JPY	112.61
GBP	1.2484
CAD	1.3023
AUD	.7680
NZD	.7315
CHF	.9930
NOK	8.2066
SEK	8.7648

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Cross Rates

Exchange rates between two non-USD currencies are **cross rates**, derived from their USD quotes.

- A cross rate is an exchange rate **between two non-USD currencies**.
- GBP per Euro (i.e. $\frac{GBP}{EUR}$) means “how many GBP we can get with one Euro?”. Euro is reference currency.
 - We first need to see how many USD a Euro is worth. That is a USD per Euro quote, i.e. $\frac{USD}{EUR}$.
 - Then with the USD per Euro quote, we see how much GBP we can get with one USD, using the $\frac{GBP}{USD}$ quote.
- Together we get, $\frac{GBP}{EUR} = \frac{USD}{EUR} \times \frac{GBP}{USD}$.
- E.g. if USD per EUR=1.1916, and USD per CAD=0.8614, what is CAD per EUR?
 - $\frac{CAD}{EUR} = \frac{USD}{EUR} \times \frac{CAD}{USD} = 1.1916 \times \frac{1}{0.8614} = 1.3833$

Bid-Ask Spread

Usually no explicit FX commission — but there is a bid-ask spread (e.g., GBP per USD: 0.8395 bid / 0.8396 ask). **Bid and ask are from the dealer's perspective:**

- **Bid** 0.8395: the dealer *buys* USD (the reference currency) at this price — the price you get when selling
- **Ask** 0.8396: the dealer *sells* USD at this price — the price you pay when buying
- We can change the reference currency by taking reciprocal
- Since buying USD is equivalent to selling GBP at $(1/0.8395) = 1.1912$ USD per GBP^a, which is the ask price for GBP.
- $\frac{1}{\left(\frac{GBP}{USD}\right)^a} = \left(\frac{USD}{GBP}\right)^b = \frac{1}{0.8396} = 1.1910$ dealer buy GBP at USD 1.1910
- $\frac{1}{\left(\frac{GBP}{USD}\right)^b} = \left(\frac{USD}{GBP}\right)^a = \frac{1}{0.8395} = 1.1912$ dealer sell GBP at USD 1.1912
- Remember dealer always buy low and sell high!

Technology and the Future of FX Markets

Electronic trading has compressed bid–ask spreads (price efficiency), democratized access (buy-side firms stream quotes), and enabled sophisticated execution algorithms.

Era	Dominant structure	Key innovation
Pre-1990s	Voice brokers, bilateral dealer calls	Reuters terminal
1990s	Electronic platforms emerge	EBS, Reuters 2000
2000s	ECNs proliferate; HFT begins	FXall, 360T, co-location
2010s	Multi-dealer platforms dominate	Algorithmic execution suites
2020s	AI-integrated, real-time analytics	LLM sentiment, ML execution

The Importance of Hedging FX Risk

Today's companies operate globally; FX movements affect profitability and performance of domestic firms. **Boeing example:** sells planes to a Brazilian buyer for \$15 million when the dollar is worth 2 reals.

- Cost to the buyer: $\$15\text{m} \times 2 = 30$ million reals
- If the dollar strengthens to 2.5 reals: $\$15\text{m} \times 2.5 = 37.5$ million reals — the buyer may switch to Airbus and pay in euros

How internationally active firms hedge: derivatives (swaps, futures); borrowing in the same currency in which they earn revenues; locating production facilities (and incurring local-currency costs) where they earn revenues.

Foreign Exchange Risk: FI Example

A U.S. FI holds: **Assets** — \$100m 1-yr USD loan at 9% + £62.5m 1-yr UK loan at 15%;
Liabilities — \$200m 1-yr USD deposits at 8%. Spot rate: \$1.6/£ (£62.5m = \$100m).

- **If the spot rate is unchanged:** UK loan returns $£62.5(1.15) \times 1.6 = \115m (15%); net return = $0.5(9\%) + 0.5(15\%) - 8\% = 4\%$
- **If GBP depreciates to \$1.45/£:** UK loan worth $£62.5(1.15) \times 1.45 = \104.22m (4.22%); net return = $0.5(9\%) + 0.5(4.22\%) - 8\% = -1.39\%$

Hedging On the Balance Sheet

On-balance-sheet hedging matches foreign assets and liabilities — FX-driven decreases in asset values are offset by decreases in liability values (and vice versa). **Example (continued):** fund the £62.5m UK loan with equivalent 1-yr pound deposits at 11%.

- No FX move: net return = $(0.5 \cdot 9\% + 0.5 \cdot 15\%) - (0.5 \cdot 8\% + 0.5 \cdot 11\%) = 2.5\%$
- GBP falls to \$1.45: UK assets yield 4.22%; UK liability costs $£62.5(1.11) \times 1.45 = \100.59m (only 0.59%)
- Net return
= $(0.5 \cdot 9\% + 0.5 \cdot 4.22\%) - (0.5 \cdot 8\% + 0.5 \cdot 0.59\%) = 6.61\% - 4.295\% = \mathbf{2.315\%}$ — similar to the no-movement case

Hedging Off the Balance Sheet

Off-balance-sheet hedging uses *forward contracts*: rates locked in at $t = 0$ apply at $t = 1$ regardless of the prevailing spot. **Example (continued)**: the FI sells the expected UK proceeds forward at \$1.55/£ for 1-yr delivery.

- At year-end it delivers £62.5(1.15)m and receives $£62.5 \times 1.15 \times 1.55 = \111.406m — a guaranteed 11.406% dollar return
- Net return = $0.5(9\%) + 0.5(11.406\%) - 8\% = \mathbf{2.203\%}$ — hedged at low cost

How Do Exchange Rates Move?

Theories describing how exchange rates *should* move:

- Purchasing Power Parity (PPP) and Relative PPP
- Interest Rate Parity (IRP)

Most parity conditions rely on the **law of one price**, which must hold in a no-arbitrage equilibrium — arbitrage is the simultaneous buying and selling of assets for a risk-free profit.

But how do rates *actually* move? Does uncovered interest parity hold? If not, the **carry trade** can profit from it.

Absolute Purchasing Power Parity

The exchange rate between two currencies equals the ratio of the two countries' price levels for an identical basket of goods:

$$S_{\text{USD/EUR}} = \frac{P_{\text{US}}}{P_{\text{Europe}}}$$

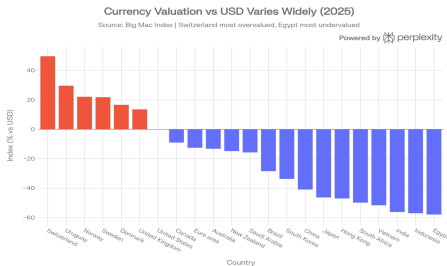
Example: a basket costs \$100 in the U.S. and EUR 90 in Europe:

$$S = \frac{100}{90} = 1.1111 \text{ USD per EUR}$$

Big Mac Currencies: PPP in Practice

At the end-2025 reading, a Big Mac cost \$5.99 in the U.S. and 7.30 francs in Switzerland (the index is updated each January and July — treat the numbers as illustrative).

- PPP-implied rate: $5.99/7.30 = 0.82$ USD per CHF; actual rate: 1.29 USD per CHF
- Swiss Big Mac in USD terms: \$9.42 — too expensive by $(9.42/5.99) - 1 = 57\%$
- For PPP to hold, Swiss prices in USD must fall 57% (the CHF must depreciate)



The Big Mac index (*The Economist*, since 1986), 2025: over/undervaluation vs. USD

Relative Purchasing Power Parity

Absolute PPP rarely holds; **relative PPP** predicts the *rate of change*:

$$\% \Delta S_{\text{USD/EUR}} = \pi_{\text{US}} - \pi_{\text{EUR}}$$

The currency of the lower-inflation country **appreciates** over time.

Example: if U.S. inflation is 5% and Euro-area inflation is 3%, the dollar should depreciate by roughly $5\% - 3\% = 2\%$ per year against the euro.

Interest Rate Parity (IRP) and Forward Contracts

Forward contract Locks in a rate today to transact foreign currency in the future; settlement on a future date by maturity of the contract (commonly 1 week, 1M, 3M). *Spot*: transact today, settle $T+2$

IRP is an **arbitrage condition** holding when currency and money markets are in equilibrium. Two one-year strategies must yield the same (riskless) return:

- 1 Invest \$1 in the U.S. at $r_{\$}$ p.a.
- 2 Convert \$1 to foreign currency at S , invest at r_{FC} p.a., and sell the proceeds *forward* at F

Forward contracts completely hedge the risk of future currency movements.

Interest Rate Parity: Derivation

Strategy 1 (invest domestically): $\text{return} = 1 + r_D$. Strategy 2 (invest abroad, covered): convert at S , invest at r_F , convert back at the forward rate:

$$\frac{1}{S} \times (1 + r_F) \times F = \frac{F}{S}(1 + r_F)$$

Both are riskless, so they must be equal:

$$1 + r_D = \frac{F}{S}(1 + r_F) \quad \implies \quad F = S \times \frac{1 + r_D}{1 + r_F}$$

The forward rate exactly offsets the interest differential — the currency with the higher interest rate trades at a forward **discount**.

Covered Interest Arbitrage (CIA)

Compare the **domestic return** $1 + r_D$ with the **foreign covered return** $\frac{F}{S}(1 + r_F)$:

- If $1 + r_D > \frac{F}{S}(1 + r_F)$: the domestic currency is overvalued in the forward market — invest domestically
- If $\frac{F}{S}(1 + r_F) > 1 + r_D$ (e.g., a 5% U.S. rate vs. a 7% German rate not offset by the forward premium): borrow domestic, convert, invest foreign, sell forward — invest in the foreign country

How CIA corrects mispricing: arbitrage flows bid up the spot rate S (or push down F) until

$$\frac{F}{S}(1 + r_F) = 1 + r_D \quad \text{— IRP restored.}$$

Covered Interest Arbitrage: Example

IRP does not hold: borrow USD at 2.405%, convert to EUR, invest EUR at 1.36%, sell EUR 6-month forward.

- Borrow \$1m at 2.405% (6-mo)
- Convert at spot \$1.1939/EUR $\rightarrow$ EUR 837,591.09
- Invest at 1.36% $\rightarrow$ EUR 848,982.33 in 6 months
- Sell 6-month forward at \$1.2072/EUR $\rightarrow$ \$1,024,891.47
- Repay \$1,024,050 $\Rightarrow$ **risk-free profit \$841.47**

If the spot rate is 1.1939 USD per EUR, 6-month forward rate is 1.2072 USD per EUR, USD 6-month interest rate is 4.81% p.a., and EUR 6-month interest rate is $r_e=2.72\%$ p.a.. Does IRP hold? Use a n -mths/12 interest accrual basis.

- First, compute 6-month rates $r_{USD}=4.81\%/2=2.405\%$; $r_{EUR}=2.72\%/2=1.36\%$. Then check the IRP relation:

$$(1 + r_{USD}) = \frac{F}{S} (1 + r_{EUR})?$$
$$1.02405 < \frac{1.2072}{1.1939} (1 + 0.0136)$$
$$1.02405 < 1.02489$$

Three Exchange Rate Regimes

The IMF classifies arrangements by *de facto* behavior, which often differs from the announced regime.

- Hard pegs** No separate legal tender (dollarization: Ecuador, El Salvador; monetary union: Eurozone) or currency board — domestic currency fully backed by reserves at a fixed rate (Hong Kong's Linked Exchange Rate System: HKD 7.80/USD since 1983). Maximum credibility, no FX risk for trade — but loss of monetary independence and lender-of-last-resort function
- Soft pegs** Conventional peg ($\pm 1\%$ of a central rate, e.g., Gulf states), crawling peg (adjusted at a predetermined rate to control inflation), stabilized arrangements. Some flexibility — but vulnerable to speculative attack when fundamentals diverge
- Floating** Managed float (discretionary intervention based on payments, reserves, market conditions) and freely floating (market-determined; intervention only to moderate excessive volatility)

The Impossible Trinity (Trilemma)

The Mundell–Fleming trilemma: a country cannot simultaneously maintain all three of **a fixed exchange rate**, **free capital mobility**, and **independent monetary policy**. Only two can coexist:

- **Eurozone:** fixed rates + free capital → surrendered monetary policy
- **Hong Kong SAR:** fixed rate + free capital → no independent monetary policy; HIBOR tracks the Fed Funds Rate
- **China (historically):** fixed rate + monetary independence → imposed capital controls
- **U.S./UK/Japan:** free float + free capital → full monetary independence

Invest in High-Interest-Rate Currencies?

All parity conditions — relative PPP, IRP — predict that high-interest-rate currencies *depreciate*.

But most investors — and most financial press coverage — assume high interest rates are *good* for a currency:

Dollar climbs after Fed lifts rates, flags more hikes

Published 11:16 PM ET Tue, 25 Sept 2018 | Updated 5:32 PM ET Wed, 26 Sept 2018



The president has also previously stated his preference for a weaker dollar, which makes US exports more competitive. The Fed's rate tightening and expectations for further rate rises on the horizon has been a key factor in boosting the buck this year. Expectations for further dollar strengthening have also increased in recent weeks.

Trump's Fed criticism sends US dollar lower

Currency faces 'serious headwind' after president says 'not thrilled' with rate rises

Uncovered Interest Rate Parity (UIRP)

If investors are indifferent to risk, the *expected* change in the spot rate offsets the interest differential — no forward cover needed:

$$\frac{E[S_T]}{S} = \frac{1 + r_D}{1 + r_F} \quad \Longleftrightarrow \quad E[\% \Delta S] \approx r_D - r_F$$

High-interest-rate currencies are *expected* to depreciate.

The Currency Carry Trade

Hence investors like the **carry trade**: borrow low-interest currencies (e.g., JPY), invest in high-interest currencies (e.g., AUD), and *do not hedge*.

- If AUD depreciates (as PPP/IRP predict): you lose the additional AUD interest earned
- If AUD does not depreciate: you earn the interest differential
- Better yet, if AUD appreciates: you earn both the differential and the appreciation

Many academic studies (e.g., Lustig & Verdelhan, 2007) show the carry trade earns **positive returns** — weird, since parity conditions say high-rate currencies should depreciate.

Average Returns from Carry Trade

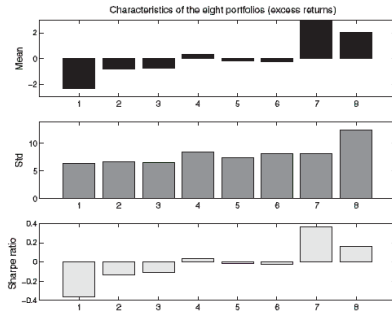


FIGURE 1. EIGHT CURRENCY PORTFOLIOS

Notes: This figure presents means, standard deviations (in percentages), and Sharpe ratios of real excess returns on eight annually rebalanced currency portfolios for a US investor. The data are annual and the sample is 1953–2002. These portfolios were constructed by sorting currencies into eight groups at time t based on the nominal interest rate differential with the home country at the end of period $t - 1$. Portfolio 1 contains currencies with the lowest interest rates. Portfolio 8 contains currencies with the highest interest rates.

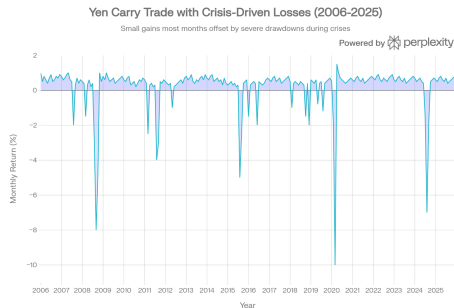
Source: Lustig & Verdelhan (*AER* 2007)

Why Does the Carry Trade Work?

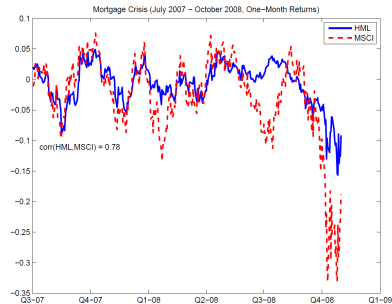
No one has a complete explanation, but suggested reasons:

- Equity risk** Carry-portfolio risk is related to equity risk: Lustig, Roussanov & Verdelhan (2012) show high-interest currencies' returns are positively correlated with equity returns
- Crash risk** Carry earns small positive returns most of the time but occasionally suffers large, sudden drawdowns. Carry tends to lose in “risk-off” episodes when volatility (VIX) spikes — investors reduce leverage and unwind positions at the same time

Carry Trade in Bad Times



Yen-funded FX carry trade, 2006–2026



HML = cumulative carry return; MSCI = U.S. stocks

From Lustig, Roussanov & Verdelhan (2012)

Safe-Haven Currencies and Global Risk Sentiment

A **safe-haven currency** appreciates (or holds value) during periods of heightened global risk aversion.

USD Primary safe haven: flight to dollar liquidity; reserve status creates structural dollar demand in stress

JPY Appreciates in risk-off as carry trades unwind and Japanese investors repatriate

CHF Current-account surplus, political neutrality, liquid capital markets

Risk-on ($VIX < 15$): investors sell JPY/CHF, buy higher-yielding commodity currencies (AUD, NZD, CAD) and EM currencies; carry performs well.

Risk-off ($VIX > 25-30$): rapid flows into USD and JPY; carry unwinds; large negative tail returns.

Other Currency Strategies

Currency momentum Buy past winners, sell past losers — a large, significant cross-sectional spread of up to 10% p.a. between winner and loser portfolios (1976–2010, 40+ currencies); not highly correlated with carry — diversification

Currency value Buy undervalued currencies (trading below PPP), sell overvalued ones. Returns lower and more variable than carry, but value diversifies against carry crash risk

Combining carry, value, and momentum generates substantially higher Sharpe ratios than any single strategy alone.

Cumulative Returns of Currency Investment Strategies

Risk-Return: FX Strategies vs. Traditional Asset Classes (Stylized)

Sources: Lustig & Verdelhan (2007); Menkhoff et al. (2012); BIS; academic estimates

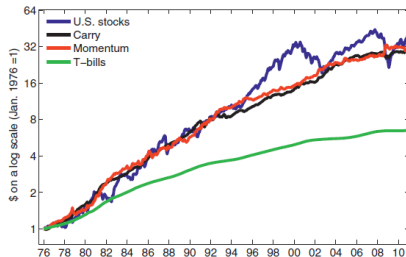
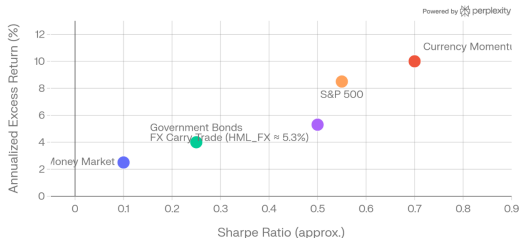


Figure 1

Cumulative monthly returns of investment strategies (Feb. 1976–Dec. 2010). The figure plots the cumulative returns of a trader who begins with \$1 in January 1976 and invests his accumulated earnings exclusively in one of four strategies. For T-bills and U.S. stocks we use the risk-free rate and value-weighted market return reported in Kenneth French's database. The carry trade and momentum portfolios are formed by taking equally weighted forward positions in up to 20 currencies versus the USD. For the carry trade portfolio, the foreign currency is sold (bought) forward if it is currently at a forward premium (discount). For the momentum portfolio, the foreign currency is sold (bought) forward, if, selling (buying) it forward in the previous period was profitable.

Key Takeaways

FX market Largest financial market (\$9.6T daily): currency conversion, exchange-rate setting, hedging tools; OTC, 24-hour global trading

Key terminology Spot (immediate exchange); forward (future exchange at a pre-specified rate); cross rates (between non-USD currencies); bid–ask spread (dealer buys at bid, sells at ask)

FX risk Currency fluctuations affect firm value; hedge on the balance sheet (match foreign assets and liabilities) or off it (forwards to lock in future rates)

PPP Exchange rates adjust to equalize prices of identical goods across countries (Big Mac index)

IRP Forward rates must offset interest differentials — otherwise covered interest arbitrage

Carry trade anomaly Borrow low-rate (JPY), invest high-rate (AUD), unhedged: violates UIP; positive average returns but crash risk (large losses in risk-off episodes / VIX spikes)